Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Validation and Scale-Up of Robust Electrocatalysts for AEM Electrolysis under Variable Water Qualities*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Chemical Processes*	*Catalyst Application Technologies*
Concept of Proposed Technology	The proposed technology targets robust, impurity-tolerant electrocatalyst and electrode concepts for anion exchange membrane water electrolysis (AEMEL) that remain stable under varying water qualities and dynamic operation typical of renewable energy integration. By improving catalyst composition, electrode microstructure and coating architectures, the approach aims to reduce dependence on costly system-level water purification while maintaining performance and durability. Advanced catalyst materials will be integrated into scalable electrode and coating concepts and validated stepwise from laboratory scale to larger-area prototype electrodes under controlled impurity profiles and fluctuating load conditions, enabling technology maturation toward TRL 5.			
Rationale for Joint R&D Project	The project requires the combination of complementary expertise: innovative electrocatalyst development from a Korean SME and application-oriented electrode engineering and electrochemical validation from Fraunhofer IFAM. While the SME contributes advanced catalyst concepts, Fraunhofer IFAM provides scalable fabrication know-how and access to industrially relevant testing infrastructure. An international joint R&D project enables efficient bridging from material innovation to validated technology, reduces development risk, and accelerates market readiness for robust AEMEL systems relevant to global hydrogen applications.			
Key Words	1. AEM electrolysis			

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	2. Impurity-tolerant electrocatalysts 3. Electrode coating 4. Water-quality robustness 5. Scale-up validation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The objective of this project is to develop and validate robust electrocatalyst and electrode concepts for alkaline membrane water electrolysis (AEMEL) that operate reliably under variable water qualities and dynamic load conditions. The project aims to bridge electrocatalyst innovation (e.g. Core-Shell concepts) and application-oriented electrode development through complementary work between a Korean SME and Fraunhofer IFAM. By integrating optimized catalysts into scalable electrode architectures and validating them in prototype-level configurations, the project targets technology maturation to TRL 5.			
R&D Target Specifications	The project targets prototype AEMEL electrocatalyst and electrode systems capable of operating at current densities of ≥1.0 A cm^{-2} with cell voltages of ≤2.1 V at ≤80 °C under defined reference conditions. Durability targets include a voltage degradation rate of ≤100 μV h^{-1} during long-term operation and stable performance over ≥1,000 h at ≥1.0 A cm^{-2}, as well as ≤10 % performance loss after defined accelerated stress and load-cycling tests. Robustness against variable water qualities will be demonstrated by maintaining ≥85 % of baseline (DI-water) performance under defined impurity levels and achieving ≥80 % performance recovery after returning to reference water. Scalability will be verified by transferring the electrode concept to ≥100 cm^2 active area with ≤10 % performance deviation compared to small-area cells. Successful achievement of these targets through production and performance evaluation of finalized electrode prototypes will demonstrate TRL 5 maturity.			
R&D Plan	The project is organized into complementary work packages (WPs) jointly executed by the Korean SME and Fraunhofer IFAM, covering electrocatalyst development, electrode integration, validation under realistic conditions, and prototype demonstration. Responsibilities are distributed to ensure balanced contributions and strong knowledge exchange. The project concludes with a prototype-level demonstration reaching TRL 5, defined as production and performance evaluation of finalized materials, components, and systems. WP1 – Application requirements and catalyst–electrode co-design (Months 1–6, Phase 1) This work package establishes target application scenarios, operating conditions, water-quality ranges, and dynamic load profiles relevant to AEM electrolysis. The Korean SME leads the definition of catalyst design requirements and supplies initial catalyst materials, while Fraunhofer IFAM defines electrode, coating, and testing requirements. Joint activities include baseline electrochemical testing and agreement on performance and durability criteria, forming the foundation for Phase 2 development. WP2 – Advanced electrocatalyst development and optimization (Months 4–18) The Korean SME leads the synthesis and optimization of advanced electrocatalyst materials, including core–shell and impurity-tolerant concepts, tailored for variable water qualities and dynamic operation. Fraunhofer IFAM supports through detailed physicochemical and electrochemical characterization, feedback on catalyst integration into electrodes, and definition of structure–performance relationships. Iterative optimization cycles ensure compatibility with scalable electrode fabrication processes. WP3 – Electrode fabrication and coating process development (Months 7–20) Fraunhofer IFAM leads the development of scalable electrode architectures by integrating optimized catalysts from WP2 into coating, printing, and fabrication processes suitable for larger active areas. The Korean SME contributes by adapting catalyst formulations, inks, and material supply to manufacturing constraints and by evaluating cost and production feasibility. Joint optimization focuses on gas management, microstructure control, and mechanical			

	stability. WP4 – Performance and durability validation under variable water qualities (Months 12–28) Electrodes developed in WP3 are jointly evaluated under controlled variations in water quality, including defined impurity levels relevant to industrial and seawater-adjacent conditions. Fraunhofer IFAM conducts long-term and accelerated stress testing, dynamic load testing, and degradation analysis, while the SME supports interpretation of results and further catalyst refinement. This WP provides critical feedback for final material and electrode design freeze. WP5 – Prototype production, scale-up, and TRL 5 demonstration (Months 24–36) In the final work package, finalized catalyst and electrode concepts are produced as prototypes and evaluated in larger-area, stack-relevant single-cell or short-stack configurations. The Korean SME ensures reproducible catalyst production at prototype scale, while Fraunhofer IFAM leads prototype electrode fabrication and performance evaluation under representative operating conditions. The project concludes with a TRL 5 milestone, demonstrating stable performance and durability of finalized materials and components in a relevant electrolysis environment. Final Project Milestone (End of Month 36): M1 – TRL 5 Demonstration: Production and performance evaluation of prototype AEMEL electrocatalysts and electrodes under variable water qualities and dynamic operation, confirming readiness for further scale-up and system-level integration.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IFAM will act as the applied research partner responsible for electrode engineering, scalable coating and fabrication, and application-oriented electrochemical validation. This includes integration of SME-developed electrocatalysts into advanced electrode and coating architectures, development of scalable fabrication routes, and comprehensive testing under representative AEMEL operating conditions, including variable water qualities and dynamic load profiles. Fraunhofer IFAM will lead durability testing (long-term and accelerated stress tests), scale-up from laboratory cells to stack-relevant larger-area prototype electrodes, and performance evaluation to demonstrate TRL 5 maturity, while providing systematic feedback to guide iterative catalyst and electrode optimization in close collaboration with the Korean SME.
Expected R&R of Korean SME Partner	The Korean SME will lead the development, optimization, and supply of advanced electrocatalyst materials tailored for AEM electrolysis, with a focus on impurity tolerance, durability, and compatibility with scalable electrode fabrication. This includes catalyst synthesis and formulation, iterative improvement based on electrochemical feedback, and adaptation of the catalyst production toward reproducible prototype batches. The SME will contribute to electrode co-design, interpretation of performance and degradation results, and assessment of manufacturability and cost aspects, ensuring that the developed catalyst and electrode concepts are suitable for prototype validation and future commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver TRL 5 prototype electrocatalyst and electrode components for anion exchange membrane water electrolysis (AEMEL), validated under variable water qualities and dynamic operating conditions. Key technological outcomes include impurity-tolerant catalyst concepts, scalable electrode coating and architecture solutions, and verified durability and performance trends from laboratory scale to larger-area, stack-relevant cell formats. These results support reduced balance-of-plant complexity and lower dependence on system-level water purification, improving the feasibility of renewable-energy-driven hydrogen production. From an industrial and economic perspective, the project will strengthen the commercialization readiness of the Korean SME's catalyst technology by providing application-oriented validation, prototype performance data, and clear benchmarks relevant to global markets. The joint work is expected to reduce development risk, enable follow-up industrial demonstrations, and foster longer-term German–Korean cooperation in hydrogen technology value chains, contributing to industrial competitiveness and export potential in electrolysis materials and components.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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